



Aror University of Art, Architecture, Design & Heritage Sukkur.

Department of Artificial Intelligence and Multimedia Gaming

Programming for AI (Spring-2026)

LAB No. 01

Prepared by: Abdul Haseeb Shaikh

Objective of Lab No. 01:

After performing lab 1, students will be able to:

- Understand and apply print() and input() functions
- Understand and apply Variables and Data types
- Understand and apply arithmetic, relational and logical operators
- Use Above concepts to solve real world problems

Lab Tasks:

Task 1: Interactive Menu Display

Objective: Practice print() with sep and end

Task:

Create a menu for a university cafeteria:

1. Tea	Rs. 50
2. Coffee	Rs. 80
3. Sandwich	Rs. 120
4. Juice	Rs. 100



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- Use **one print statement** for the menu
- Use sep creatively

Task 2: Student Information Formatter

Objective: Input handling + formatted output

Task:

Take input:

- Name
- Roll Number
- Semester
- GPA

Display output like:

```
Name | Roll No | Semester | GPA
Ali | 23-AI-17 | 4th | 3.45
```

Note: Must use sep

Task 03: Salary Breakdown Calculator

Objective: Arithmetic operators

Task:

Input:

- Basic salary

Calculate:

- House Rent = 30%
- Medical Allowance = 10%
- Gross Salary = sum of all



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Output Format:

```
Basic Salary      : 50000
House Rent (30%)  : 15000
Medical (10%)     : 5000
Gross Salary      : 70000
```

Task 4: Logical Eligibility Checker

Objective: Relational + Logical operators

Task:

Input:

- Age
- Intermediate Marks

Check and display result for **Admission Eligibility** using logic:

If the age is above 18 and Marks are 60 or above then the output should be as follows:

```
Eligible for Admission: True
```

Task 5: Exam Result Analyzer

Objective: Relational operators

Task:

Input:

- Total marks
- Obtained marks

Display:



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- Percentage
- Boolean result for:
 - Passed ($\geq 50\%$)
 - Distinction ($\geq 80\%$)

Example Output:

```
Percentage: 82.5  
Passed: True  
Distinction: True
```

Task 06: Write a Python program that:

1. Takes student information
2. Takes marks of 3 subjects
3. Calculates:
 - Total Marks
 - Percentage
4. Displays:
 - Pass/Fail status
 - Distinction status
5. Prints a properly formatted **Result Card**



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RESULT CARD	
Name	: Ali Ahmed
Roll No	: AI-23-17
Marks:	
Subject 1	: 75
Subject 2	: 82
Subject 3	: 90
Total Marks: 247 / 300	
Percentage : 82.33	
Passed	: True
Distinction:	True